

Conditional Moment Matching and a Log-Free Multivariate Martingale Central Limit Theorem

A Rigorous Restricted Result on a Question of Srikant

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Abstract

We give a self-contained proof of an $O(n^{-1/2})$ multivariate 1-Wasserstein bound for normalized sums of martingale differences under conditional moment matching through order three and a uniform fourth-moment bound. More generally, we obtain an explicit nonasymptotic estimate in terms of weighted discrepancies of the predictable covariance and conditional third-moment tensor. The proof is a Gaussian-smoothed Lindeberg replacement carried to third order: cancellation of the cubic term replaces a harmonic remainder by the summable series $\sum_j j^{-3/2}$.

The mathematical result is valid, but its status relative to the motivating question must be stated carefully. It completely solves a restricted conditional-moment-matching subproblem, while giving only narrow partial progress on Srikant's broader question about removing the logarithm for martingales and Markov chains. For stationary Poisson-equation martingales, the conditions below ordinarily force statewise covariance matching and statewise vanishing of the conditional third moment, and hence do not cover a generic geometrically ergodic Markov chain. A substantially broader $O(n^{-1/2})$ Markov-chain result was subsequently obtained by Zhang and Xie.

1 Introduction and status of the result

Srikant proves $O((\log n)/\sqrt{n})$ bounds in multivariate 1-Wasserstein distance for martingale and Markov-chain central limit theorems, and asks whether the logarithm can be removed, possibly under additional assumptions [2]. The question is deliberately open-ended. Thus, a theorem under additional assumptions should not be called a solution without explaining how restrictive those assumptions are.

Status. The theorem and proof in this note are mathematically sound. They give a complete $O(n^{-1/2})$ result for a natural but restrictive class of martingale differences whose first three conditional moments match those of a Gaussian increment, together with a weighted perturbation version. Relative to Srikant's broader problem, this is *partial progress, not a full solution*. In particular, it does not by itself remove the logarithm for the generic Markov-chain martingales considered in Srikant's paper.

The main proof is elementary once the correct smoothing scale is exposed. A Lipschitz test function smoothed by Gaussian variance t has an r th derivative of order $t^{-(r-1)/2}$. A second-order expansion therefore leaves a cubic error proportional to

$$\frac{1}{\sqrt{n}} \frac{1}{n-k},$$

whose sum over the replacement index k is of order $(\log n)/\sqrt{n}$. If the conditional third-moment tensor vanishes, one can expand through degree three. The fourth-order error is proportional to

$$\frac{1}{\sqrt{n}} \frac{1}{(n-k)^{3/2}},$$

and this is summable.

The argument belongs to the Lindeberg–Stein tradition for quantitative martingale central limit theorems; see, for example, Röllin [1]. No claim of priority is made here. The purpose of this note is to state the restricted theorem precisely, prove it without gaps, and distinguish it from the broader Markov-chain problem. For that broader direction, Zhang and Xie [3] prove an $O(n^{-1/2})$ W_1 bound for geometrically ergodic Markov chains under substantially broader moment and regularity conditions.

2 Notation and main theorem

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration $(\mathcal{F}_k)_{k \geq 0}$. Let $(m_k)_{k \geq 1}$ be an $\mathbb{R}^d$ -valued martingale difference sequence, meaning that m_k is $\mathcal{F}_k$ -measurable, $\mathbb{E} \|m_k\| < \infty$, and

$$\mathbb{E}[m_k \mid \mathcal{F}_{k-1}] = 0 \quad \text{almost surely.}$$

For integrable random vectors Y, Y' in $\mathbb{R}^d$, define

$$W_1(Y, Y') := \sup_{h: \text{Lip}(h) \leq 1} |\mathbb{E}h(Y) - \mathbb{E}h(Y')|.$$

This is the Kantorovich–Rubinstein representation of the 1-Wasserstein distance between their laws.

For an order- r tensor $A = (A_{i_1 \dots i_r})$, write

$$\|A\|_F := \left(\sum_{i_1, \dots, i_r=1}^d A_{i_1 \dots i_r}^2 \right)^{1/2}.$$

For two tensors of the same order, $\langle A, B \rangle$ denotes their coordinatewise inner product. If $u \in \mathbb{R}^d$, then $\|u^{\otimes r}\|_F = \|u\|^r$.

Fix a positive definite matrix $\Sigma \in \mathbb{R}^{d \times d}$ and define the whitened increments

$$X_k := \Sigma^{-1/2} m_k, \quad V_k := \mathbb{E}[X_k X_k^\top \mid \mathcal{F}_{k-1}], \quad T_k := \mathbb{E}[X_k^{\otimes 3} \mid \mathcal{F}_{k-1}].$$

For $n \geq 2$, let

$$A_n := \sum_{k=1}^{n-1} \frac{\mathbb{E} \|V_k - I_d\|_F}{\sqrt{n-k}},$$

$$B_n := \sum_{k=1}^{n-1} \frac{\mathbb{E} \|T_k\|_F}{n-k}.$$

Theorem 2.1 (Nonasymptotic log-free criterion). *Let $Z \sim \mathcal{N}(0, I_d)$ and suppose*

$$M_4 := \sup_{k \geq 1} \mathbb{E} \|X_k\|^4 < \infty.$$

There are finite constants $\kappa_{2,d}, \kappa_{3,d}, \kappa_{4,d}$, depending only on d , such that for every $n \geq 2$,

$$\begin{aligned} & W_1 \left(\frac{1}{\sqrt{n}} \sum_{k=1}^n m_k, \Sigma^{1/2} Z \right) \\ & \leq \frac{\|\Sigma^{1/2}\|_{\text{op}}}{\sqrt{n}} \left[M_4^{1/4} + \sqrt{d} + \frac{\kappa_{2,d}}{2} A_n + \frac{\kappa_{3,d}}{6} B_n \right. \\ & \quad \left. + \frac{\kappa_{4,d}}{24} (M_4 + d(d+2)) \sum_{j=1}^{n-1} j^{-3/2} \right]. \end{aligned} \quad (1)$$

Consequently, if

$$\sup_{n \geq 2} A_n < \infty \quad \text{and} \quad \sup_{n \geq 2} B_n < \infty,$$

then

$$W_1 \left(\frac{1}{\sqrt{n}} \sum_{k=1}^n m_k, \Sigma^{1/2} Z \right) = O(n^{-1/2}).$$

A sufficient, but generally stronger, condition is

$$\sum_{k=1}^{\infty} \mathbb{E} \|V_k - I_d\|_{\text{F}} < \infty, \quad \sum_{k=1}^{\infty} \mathbb{E} \|T_k\|_{\text{F}} < \infty.$$

The cleanest case is exact conditional moment matching.

Corollary 2.2 (Matching through order three). *Suppose, almost surely for every k ,*

$$\mathbb{E}[X_k X_k^{\top} \mid \mathcal{F}_{k-1}] = I_d, \quad \mathbb{E}[X_k^{\otimes 3} \mid \mathcal{F}_{k-1}] = 0,$$

and $\sup_k \mathbb{E} \|X_k\|^4 = M_4 < \infty$. Then

$$\begin{aligned} W_1 \left(\frac{1}{\sqrt{n}} \sum_{k=1}^n m_k, \Sigma^{1/2} Z \right) & \leq \frac{\|\Sigma^{1/2}\|_{\text{op}}}{\sqrt{n}} \left[M_4^{1/4} + \sqrt{d} \right. \\ & \quad \left. + \frac{\kappa_{4,d}}{24} (M_4 + d(d+2)) \zeta(3/2) \right]. \end{aligned} \quad (2)$$

Corollary 2.3 (Conditional symmetry). *Suppose*

$$\mathbb{E}[m_k m_k^{\top} \mid \mathcal{F}_{k-1}] = \Sigma \quad \text{almost surely}$$

and

$$\mathcal{L}(m_k \mid \mathcal{F}_{k-1}) = \mathcal{L}(-m_k \mid \mathcal{F}_{k-1}) \quad \text{almost surely}$$

for every k . If $\sup_k \mathbb{E} \left\| \Sigma^{-1/2} m_k \right\|^4 < \infty$, then the bound (2) holds.

Proof. Conditional symmetry makes every integrable odd conditional moment vanish, in particular $T_k = 0$. The covariance assumption gives $V_k = I_d$. Apply Corollary 2.2. $\square$

3 Gaussian smoothing

For a measurable function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ of at most linear growth and $t > 0$, define the heat semigroup

$$P_t h(x) := \mathbb{E} h(x + \sqrt{t} Z), \quad Z \sim \mathcal{N}(0, I_d).$$

Let ϕ_d be the standard Gaussian density. The order- r probabilists' Hermite tensor $\mathbf{H}_r$ is defined by

$$\mathbf{H}_r(z) \phi_d(z) = (-1)^r D^r \phi_d(z).$$

Lemma 3.1 (Derivative bounds for Lipschitz functions). *For every integer $r \geq 1$, every 1-Lipschitz h , and every $t > 0$, $P_t h$ is smooth and*

$$\sup_{x \in \mathbb{R}^d} \|D^r P_t h(x)\|_F \leq \kappa_{r,d} t^{-(r-1)/2},$$

where one may take

$$\kappa_{r,d} := \mathbb{E}[\|Z\| \|\mathbf{H}_r(Z)\|_F] < \infty.$$

Proof. A Lipschitz function has at most linear growth, while every derivative of a Gaussian density is a polynomial times that density. Hence differentiation under the convolution integral is justified by dominated convergence. A change of variables gives

$$D^r P_t h(x) = t^{-r/2} \mathbb{E}[h(x + \sqrt{t} Z) \mathbf{H}_r(Z)].$$

Since $\mathbb{E} \mathbf{H}_r(Z) = 0$, we may subtract $h(x)$ inside the expectation. Therefore

$$\begin{aligned} \|D^r P_t h(x)\|_F &\leq t^{-r/2} \mathbb{E} \left[\left| h(x + \sqrt{t} Z) - h(x) \right| \|\mathbf{H}_r(Z)\|_F \right] \\ &\leq t^{-(r-1)/2} \mathbb{E}[\|Z\| \|\mathbf{H}_r(Z)\|_F]. \end{aligned}$$

The last expectation is finite because $\mathbf{H}_r$ is polynomial and the Gaussian distribution has moments of every order. $\square$

4 Proof of Theorem 2.1

We first work in whitened coordinates. Set

$$\bar{S}_n := \frac{1}{\sqrt{n}} \sum_{k=1}^n X_k.$$

Let $Z_1, \dots, Z_n$ be independent $\mathcal{N}(0, I_d)$ random vectors, independent of $\mathcal{F}_\infty := \sigma(\cup_{k \geq 0} \mathcal{F}_k)$. For a 1-Lipschitz function h , define the Lindeberg hybrids

$$Y_k := \frac{1}{\sqrt{n}} \left(\sum_{i=1}^k X_i + \sum_{i=k+1}^n Z_i \right), \quad k = 0, \dots, n.$$

Then $Y_0 \stackrel{d}{=} Z$ and $Y_n = \bar{S}_n$, so

$$\mathbb{E} h(\bar{S}_n) - \mathbb{E} h(Z) = \sum_{k=1}^n \Delta_k, \quad \Delta_k := \mathbb{E}[h(Y_k) - h(Y_{k-1})].$$

Fix $k \leq n-1$ and write

$$U_k := \frac{1}{\sqrt{n}} \sum_{i=1}^{k-1} X_i, \quad t_k := \frac{n-k}{n}, \quad H_k := P_{t_k} h.$$

After integrating out $Z_{k+1}, \dots, Z_n$,

$$\Delta_k = \mathbb{E} \left[H_k \left(U_k + \frac{X_k}{\sqrt{n}} \right) - H_k \left(U_k + \frac{Z_k}{\sqrt{n}} \right) \right].$$

For a C^4 function f , Taylor's formula along the line segment from x to $x+y$ gives

$$\begin{aligned} f(x+y) &= f(x) + Df(x)[y] + \frac{1}{2} D^2 f(x)[y^{\otimes 2}] + \frac{1}{6} D^3 f(x)[y^{\otimes 3}] + R_4(f; x, y), \\ |R_4(f; x, y)| &\leq \frac{1}{24} \sup_{z \in \mathbb{R}^d} \|D^4 f(z)\|_{\text{F}} \|y\|^4. \end{aligned}$$

Condition on $\mathcal{F}_{k-1}$. The vector U_k is $\mathcal{F}_{k-1}$ -measurable, $\mathbb{E}[X_k | \mathcal{F}_{k-1}] = 0$, and Z_k is an independent standard Gaussian. Hence the constant and linear terms cancel, while $\mathbb{E}[Z_k^{\otimes 3}] = 0$. It follows that

$$\Delta_k = \frac{1}{2n} \mathbb{E} \langle D^2 H_k(U_k), V_k - I_d \rangle + \frac{1}{6n^{3/2}} \mathbb{E} \langle D^3 H_k(U_k), T_k \rangle + \rho_k, \quad (3)$$

where

$$|\rho_k| \leq \frac{1}{24n^2} \|D^4 H_k\|_{\infty, \text{F}} (\mathbb{E} \|X_k\|^4 + \mathbb{E} \|Z_k\|^4). \quad (4)$$

Here $\|D^r f\|_{\infty, \text{F}} := \sup_x \|D^r f(x)\|_{\text{F}}$. By Cauchy-Schwarz for the tensor inner product, Lemma 3.1, and $\mathbb{E} \|Z_k\|^4 = d(d+2)$,

$$\begin{aligned} |\Delta_k| &\leq \frac{\kappa_{2,d}}{2\sqrt{n}} \frac{\mathbb{E} \|V_k - I_d\|_{\text{F}}}{\sqrt{n-k}} + \frac{\kappa_{3,d}}{6\sqrt{n}} \frac{\mathbb{E} \|T_k\|_{\text{F}}}{n-k} \\ &\quad + \frac{\kappa_{4,d}}{24\sqrt{n}} \frac{M_4 + d(d+2)}{(n-k)^{3/2}}. \end{aligned} \quad (5)$$

Indeed,

$$\|D^r H_k\|_{\infty, \text{F}} \leq \kappa_{r,d} \left(\frac{n}{n-k} \right)^{(r-1)/2}, \quad r = 2, 3, 4.$$

There is no remaining Gaussian smoothing at the last replacement $k = n$. The Lipschitz property alone yields

$$|\Delta_n| \leq \frac{1}{\sqrt{n}} (\mathbb{E} \|X_n\| + \mathbb{E} \|Z_n\|) \quad (6)$$

$$\leq \frac{M_4^{1/4} + \sqrt{d}}{\sqrt{n}}. \quad (7)$$

Summing (5) for $k = 1, \dots, n-1$, adding (7), and taking the supremum over all 1-Lipschitz h proves the whitened form of (1).

Finally, for every deterministic matrix A and integrable random vectors Y, Y' ,

$$\mathbf{W}_1(AY, AY') \leq \|A\|_{\text{op}} \mathbf{W}_1(Y, Y').$$

Apply this with $A = \Sigma^{1/2}$. The rate statements follow because $\sum_{j \geq 1} j^{-3/2} = \zeta(3/2) < \infty$ and, under absolute summability, A_n and B_n are uniformly bounded. This proves Theorem 2.1. $\square$

5 A genuinely dependent example

Exact conditional covariance and conditional symmetry do not imply independence.

Example 5.1 (A predictable rotation). Work in $d = 2$. Let $\varepsilon_1, \varepsilon_2, \dots$ be independent and uniform on

$$\mathcal{S} := \{\pm\sqrt{2}e_1, \pm\sqrt{2}e_2\}.$$

Let R be rotation by angle $\pi/4$, let $\mathcal{F}_k := \sigma(\varepsilon_1, \dots, \varepsilon_k)$, and define

$$Q_1 := I_2, \quad Q_2 := \begin{cases} I_2, & \varepsilon_1 \in \{\pm\sqrt{2}e_1\}, \\ R, & \varepsilon_1 \in \{\pm\sqrt{2}e_2\}, \end{cases} \quad Q_k := I_2 \quad (k \geq 3).$$

Set $X_k := Q_k \varepsilon_k$. Since Q_k is orthogonal and $\mathcal{F}_{k-1}$ -measurable,

$$\mathbb{E}[X_k \mid \mathcal{F}_{k-1}] = 0, \quad \mathbb{E}[X_k X_k^\top \mid \mathcal{F}_{k-1}] = I_2,$$

and the conditional distribution of X_k is symmetric about zero. Also $\|X_k\|^4 = 4$ almost surely. Hence Corollary 2.3 applies.

The sequence is not independent. Indeed, $R\mathcal{S}$ is disjoint from $\mathcal{S}$, and therefore

$$\mathbb{P}(X_2 \in \mathcal{S} \mid X_1 \in \{\pm\sqrt{2}e_1\}) = 1, \quad \mathbb{P}(X_2 \in \mathcal{S} \mid X_1 \in \{\pm\sqrt{2}e_2\}) = 0.$$

For any positive definite Σ , the increments $m_k := \Sigma^{1/2} X_k$ thus form a genuinely dependent martingale difference sequence satisfying the log-free theorem.

6 Why this is only partial progress on the Markov-chain problem

The exact estimate in Theorem 2.1 is nontrivial, but its covariance and third-moment criteria are much stronger than ordinary mixing. The following elementary observation makes the limitation precise.

Proposition 6.1 (Stationarity forces statewise matching). *Let $(Y_k)_{k \geq 0}$ be stationary with marginal law π . Suppose that for some measurable tensor-valued functions G and H ,*

$$V_k = G(Y_{k-1}), \quad T_k = H(Y_{k-1}) \quad \text{almost surely for every } k.$$

Then

$$A_n = \mathbb{E}_\pi \|G(Y_0) - I_d\|_F \sum_{j=1}^{n-1} j^{-1/2},$$

$$B_n = \mathbb{E}_\pi \|H(Y_0)\|_F \sum_{j=1}^{n-1} j^{-1}.$$

Consequently, $\sup_n A_n < \infty$ and $\sup_n B_n < \infty$ hold if and only if

$$G(Y_0) = I_d \quad \text{and} \quad H(Y_0) = 0 \quad \pi\text{-almost surely.}$$

Proof. Stationarity makes each expectation in the definition of A_n equal to the same constant $a := \mathbb{E}_\pi \|G(Y_0) - I_d\|_F$, and each expectation in B_n equal to $b := \mathbb{E}_\pi \|H(Y_0)\|_F$. Reindexing with $j = n - k$ gives the two displayed identities. Since both partial sums diverge, uniform boundedness forces $a = b = 0$, which is equivalent to the asserted almost-sure identities. The converse is immediate. $\square$

For the Poisson-equation martingale used in Srikant’s Markov-chain argument, the increment has the form

$$m_k = \psi(Y_k) - \mathbb{E}[\psi(Y_k) \mid Y_{k-1}],$$

where ψ is a solution of the relevant Poisson equation. Its conditional covariance and conditional third-moment tensor are functions of the current state Y_{k-1} . In stationarity, Proposition 6.1 shows that the present theorem applies only when those statewise tensors exactly match the Gaussian ones. Geometric mixing generally controls averages such as $\mathbb{E}V_k$ or the distribution of Y_k , not the pathwise discrepancy $\mathbb{E}\|V_k - I_d\|_F$ appearing here. Thus the theorem does not address the generic Markov-chain case that motivates Srikant’s question.

This distinction is not merely semantic. Zhang and Xie establish an $O(n^{-1/2})$ multivariate W_1 central limit theorem for geometrically ergodic Markov chains under substantially broader assumptions [3]. Their theorem therefore addresses the principal Markov-chain direction without requiring statewise conditional moment matching.

7 Conclusion

The third-order Gaussian-smoothed Lindeberg argument is correct and yields a clean $O(n^{-1/2})$ theorem for a genuinely dependent class of multivariate martingale differences. The mechanism is exact: conditional cancellation of the third tensor changes a harmonic error sum into a summable fourth-order error sum.

The proper classification is nevertheless narrow. This note gives a complete solution of the restricted conditional-moment-matching subproblem and a weighted perturbation thereof. It gives partial progress, rather than a full solution, on Srikant’s broader martingale and Markov-chain question. Any presentation of the result should make that distinction explicit.

References

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